graphics/Logo-schwarz.pdf

Design Games

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Sami Pasquali
Matrikelnummer 11809912

an der Fakultät für Informatik

der Technischen Universität Wien

Betreuung: Pretitle Hilda Tellioğlu, Posttitle

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Sami Pasquali	Hilda Tellioğlu

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Design Games

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Advisor: Pretitle Hilda Tellioğlu, Posttitle

Vienna, September 30, 2026

Sami Pasquali	Hilda Tellioğlu

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Sami Pasquali

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Danksagung

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Kurzfassung

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Abstract

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CHAPTER 1

Design Games

[Bra] describes Design games as a way of organizing participation in design projects. [TPT] notes that they (Design Games) provide a starting point for facilitating user participation and contribution.

1.1 History

1.2 Connection to Games

1.3 Categories and Classification of Design Games

1.4 Mechanics of Design Games

Design games are most of the time purpose-build for a specific design context [Bra]. Various frameworks have been proposed in the co-design literature to provide building blocks for the development of design games [Bra, VM]. Nonetheless, through the literature review done in this thesis a set of 22 recurring mechanics have been identified.

For further analysis, these mechanics were grouped into five clusters according to the effects they have when employed in a co-design workshop. A single mechanic can be attributed to multiple effects, depending on the lens it is viewed through. Those mechanics are grouped into more than one cluster of effects. For each mechanic a brief explanation grounded in the literature describes its connection with the effect it is attributed for.

Table 1.1 provides an overview of the identified mechanics and their respective cluster assignments.

Mechanic	Break habits	Generate ideas	Build shared understanding	Learn participant concepts	Address power relations
Abstraction		x	x	x	
Acting				x	x
Boundary objects		x	x	x	
Chance	x	x		x	x
Combining elements that do not fit together	x	x			
Competition					x
Design context		x			
Establishing an informal atmosphere		x			
Facilitation	x	x			x
Fun					x
Game rules	x	x	x	x	x
Involving stakeholders	x				
Magic circle	x				
Magic if				x	
Open-endedness		x			x
Participant concepts				x	
Presupplied input	x	x	x		
Scenarios		x			
Tangible game material	x	x	x	x	
Time keeping		x			x
Turn taking					x
Win condition		x			

Table 1.1: Identified mechanics and the cluster of effect they are attributed to. Ordered alphabetically.

1.4.1 Break existing habits

Breaking existing habits is a shift from everday logical thinking to generative, imaginative and lateral thinking. It grounds abstract thinking into a concrete reality so participants aren't overwhelmed by a blank canvas and overcomes potential power relations participants are usually part of.

Chance

A dice or randomly drawing cards might be used to select which topics or questions to discuss, easing the pressure to decide where to start and avoiding participants selecting the topics they are most comfortable with [Bra, HM, BHK].

Combining elements that do not fit together

"(..) people can be triggered out of their normal thinking patterns by, sometimes unusual, associations, suggestions or actions written and/on illustrated on cards" [RW]

Element of fun & Ludic drive

Players are absorpted in play. This is commonly referred to as the "Fun element" [Bra].

Facilitation

A facilitator can leave out elements to increase the focus more on a specific topic or influence which topic or question is to be worked on next [KKM]

Game rules

Design games establish a distinct space for creative exploration that is separate from everyday life. This space is created by the games rules, which define the objectives, criteria of success and allowed actions for the role players adopt during game playing [FGKM]. Furthermore, rules help to establish structured discussions which is shown to make it easier for participants to reach consensus [ZPZ$^+$].

Involving stakeholders

Design games involve stakeholders from different backgrounds and invite them to collaborate. By doing so existing routines are halted and out of ordinary communication is established that allows new problem solving approaches to emerge [FGKM].

Magic circle

Magic circle is a sphere of experience distinct from ordinary life. It may be achieved, for example, by a special meeting room setup. By entering the magic circle, participants think beyond their usual patterns and are invited to invision alternative solutions [VM, FGKM]

Presupplied input

How topics are framed by the supplied game material matters to the outcome of the game [BMB].

Tangible game material

Through Concept Design Games it has been found that Game Pieces and Game Boards can be abstracted to eliminate functional knowledge and former experiences participants bring with them [Bra]

1.4.2 Come up with creative ideas & Overcoming cognitive intertia

Design games create a kind of "scaffolding" to help non-designers access their creativity. Without some structure, people default to what they already know, or experience analysis paralysis.

Abstracted and ambiguous tasks

Help participants to propose new interpretations and alternative solutions to come up with creative ideas [VM], [BHK].

Boundary objects

Boundary objects make arguments tangible and help participants to focus [BM].

Chance

Elements of chance can be used to decide for players which topic or questions to be worked on next, avoiding analysis paralysis [HM]. A specific example is to randomly draw cards, which can then act as starting point for discussions or as creative input for building a scenario [BHK]. Design games are often created open-ended to allow pivots in decisions caused by unforseen events up until the very end [FGKM].

Combining elements that do not fit together

Combining elements that normally do not fit together is a technique that can be used to spark creativity [Bra]. Another approach is to violate known information by, for example, placing cards wrongly. This causes the brains creativity to repair the violation in an ad-hoc manner [RW].

Design Context

Design Games are specifically designed for a specific context in mind. Depending on this context, different game qualitites are desirable, and different participants are needed [VM], [GCAG], [FGKM].

Establishing an informal atmosphere

"Game playing creates an informal atmosphere, which is the most productive in creative work" [Bra].

Facilitation

A facilitator can lead the workshop and present the agenda, game pieces used and rules to follow [EBMV]. Facilitators can also take an active role during the workshop and drive the process forward by making suggestions, as well as encouraging players to make moves, unblocking negotiations [BM].

Game rules

Rules set the boundaries for whats possible and also structure the play of the game. The shift from the goals of the design process to the rules of a game makes it easier to generate design moves. Further, restrictions on idea generation is shown to improve creative outcome [BM].

Open-endedness

Design games foster open-ended exploration by combining ambiguous prompts [VM] and tactile artifacts—such as game cards—that invite participant annotation [HM], allowing the structure of the game itself to evolve organically through play [FGKM].

Presupplied input

Presupplied materials as Input support the creative process in various ways. Premade prompts can spark creativity and discussions [Bra, NETM$^+$], while predesigned items help in establishing the frame and context of the session [VM, GCAG]. Throughout the workshop, visual themes help guide ongoing conversations [HM]. Additionally, cards can be used to hint participants towards best practices in design or remind them of domain specifics [RW, NETM$^+$].

Scenarios

Design Games construct scenarios that are both concrete and flexible - "They represent a particular interpretation of a design situation" [BM]. Scenario construction can be used for creating and testing design ideas with future users. Generic game pieces can be provided for that or users encouraged to produce their own [Bra].

Split participants in groups

Participants may play the same game in parallel, split in groups. Thus each group comes to their own conclusion [EBMV]. Another approach is to let participants first individually create ideas to then share them with the rest of their team or between groups [BHK].

Tangible game material

In design games, visual and physical elements establish shared focus for participants while acting as continuous reference material during and after the design session [VM]. Furthermore, the ability to easily move and combine physical elements, like cards, helps participants to more easily connect different themes, offering new insights [HM].

Time keeping

Keeping track of time can improve structure of discussion and prompt the team to conclude their discussions [EBMV]. Time keeping also aids in equally distributing time

between the different topics to discuss [EBMV]. In design games players often do not compete against each other, but might collaborate to "race against the clock". [FGKM].

Win condition

There is rarely competition against each other in design game. However, tension is a quality of design games and aids the creative process [VM]. Misalignment between goals - for example an individual one for the player and an overall one for the game - can also impede the success of a design game session [KGPC, FGKM].

1.4.3 Create common understanding

Multiple participants from different backgrounds participate in co-design sessions. Design Games often allow a common language to emerge between players. Constraints imposed by the game force participants to translate their tacit knowledge into this shared language and be explicit in their moves and explanations.

Abstracted and stylized game pieces

Abstracted and stylized game pieces prompt participants to be explicit about their understanding and views [Bra], [VM], [KKM].

Boundary objects

Boundary objects make discussions tangible. They externalize arguments and concepts so that a common ground is established to which every participant can relate to. At the same time they are rich enough to close the gap between different viewpoints [BM].

Game rules

Rules serve as a bridge across disciplinary divides, establishing a common ground that all participants can understand [Bra]. Furthermore, rules dictate communication patterns, prompting direct dialogue between players to help them navigate game obstacles [ZPZ+]. When these rules are intentionally left vague, they encourage participants to openly articulate their own interpretations and perspectives [Bra].

Presupplied Input

Sometimes proposals or details about designs are supplied preprinted to the design-game session. Participants can point towards certain details in the proposal during the decision making arguments visible for everyone [EBMV].

Tangible game material

Tangible game pieces capture discussions physically for every participant to see [Bra], [EBMV]. They also help participants to track their role within negotiations. Similar to

boundary objects, they are rich enough to allow participants to interpret them differently, while at the same time bridging the gap between different disciplines and understandings [BM].

1.4.4 Learn about concepts participants hold & Build empathy

Through playing different roles players develop empathy for different perspectives. Abstracted game pieces and ambiguous tasks prompt players to make explicit the concepts and knowledge they hold.

Abstracted and stylized game pieces

Abstracted and stylized game pieces prompt participants to be explicit about their understanding and views. This allows to learn about concepts and knowledge participants hold [Bra], [VM], [KKM]

Acting

Enacting scenarios helps participants to build empathy for the experiences of the persons to design for. It also allows participants to switch roles and gain new perspectives on a topic. [BM], [VM]. The choice of scenes and what details to include further makes visible concepts participants hold [KKM]. "In performances, play and games the reactions of performers and audience are real, even though the actions that trigger these are fictional" [VM].

Boundary objects

Tangible boundary objects make visible how other participants interpret the material. [BM]

Chance

Elements of chance, for example a dice throw, can be used to, for example, select a role through their perspective participants should then discuss a story. This allows participants to empathize with this role. [KKM]

Magic if

The magic if describes the question of "if this person is in this situation - how would she react?". When enacting scenarios, actors frequently need to ask themself this question which increases empathy for the role choosen to enact [Bra].

Participant concepts

Design games do not simply generate designs or facilitate collaboration but are a research tool itself. They make explicit and challenge perspectives of both participants and

researchers. Reactions and outcome from involving stakeholders can be surprisingly unfamiliar and different from what researchers initially expect [KKM].

Rules

Participants interpret rules against the background of their own experiences and expectations, allowing those to become explicit [FGKM].

Tangible game material

Tangible game pieces allow participants to use more of their senses. They make concepts and throught processes explicit and allow more reflection [Bra]. Certain game pieces also allow participants to think through the eyes of a different role [GCAG].

1.4.5 Overcome power relations & Lower communication barriers in a group setting

Participatory design aims to include affected stakeholders in the design process and thus equalize power relations. Various game mechanics used in design games support that. Collaboratively working to win against a shared obstacle, rules structuring communication structure, turns to equalize attention among players and an informal atmosphere help to equalize power hierarchies and lower communication barriers.

Acting

Scenarios can be explored by enacting them. Having users create and enact those scenarios, while within their own environment lessens the power distance to the design team [Bra].

Agenda & Facilitation

Often game designers, researchers or a facilitor decides beforehand what issues to include in the session. This can negatively impact or manifest power relations in a hidden way [EBMV]

Chance

Elements of chance, e.g. a dice or some wheel, can be used to decide which participant to start with their turn [GCAG].

Competition

Design games usually do not include competition among participants but are often designed such that participants need to collaborate to overcome a shared obstacle, levelling power relations [FGKM].

Fun

The game like format of design games seems to balance usual power relations between participants and stakeholders [GCAG], [ZPZ+]. It also shows a lower entry barrier for participants to join discussions [NETM+].

Involve stakeholders

Participatory design tools, like design games, encourage stakeholders and users to participate, which encourages cross-disciplinary collaboration [NETM+], [ZPZ+]. Often people with different backgrounds, expertise and interests are thus included and their perspectives and interests added to the discussion [BHK].

Open-endedness

The open-endedness of assignments and tasks in design games allows participant to interpret them and thus exercise influence on what to focus on in the design [Bra].

Rules

Adhering to game rules minimizes power imbalances between participants [BM], [Bra]. Rules alter standard communication patterns by encouraging players to willingly navigate obstacles, such as following specific guidelines on when to engage in explicit discussions with stakeholders [ZPZ+].

Time keeping

Enforcing time rules allows to equally distribute time between players taking turns or between issues to discuss [EBMV]. Time can also be used as an obstacle players have to collaboratively win against when playing the game [FGKM].

Turn taking

Turn taking divides attention among players and allows participation on more equal terms [Bra], [EBMV], [HM].

1.4.6 Discussion

Table 1.1 shows that the mechanic Game rules is part of every design game analysed. The mechanics Chance, Game rules, Tangible game material appear in almost every cluster (4 out of 5). It can be argued that those are, despite the different contexts design games are usually designed for, core components of most, if not all, design games.

Mechanics Abstraction, Boundary objects,

1.5 Benefits

1.6 Problems and Issues

CHAPTER 2

Examples of Design Game and their use in different domains

CHAPTER 3

Design Games in Design Thinking

3.1 Opportunities for Design Game use

Overview of Generative AI Tools Used

No AI tools were used for creating this paper.

Übersicht verwendeter Hilfsmittel

Keine

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Bibliography

[BHK] Tina Helene Bunæs, Michelle Husebye, and Joakim Karlsen. Rules of Play to Balance Ideation and Decision Making in Co-design Games. pages 33–40.

[BM] Eva Brandt and Jörn Messeter. Facilitating collaboration through design games. In *Proceedings of the Eighth Conference on Participatory Design: Artful Integration: Interweaving Media, Materials and Practices - Volume 1*, PDC 04, pages 121–131. Association for Computing Machinery.

[BMB] Eva Brandt, Jörn Messeter, and Thomas Binder. Formatting design dialogues – games and participation. 4(1):51–64.

[Bra] Eva Brandt. Designing exploratory design games: A framework for participation in Participatory Design? In *Proceedings of the Ninth Conference on Participatory Design: Expanding Boundaries in Design - Volume 1*, PDC '06, pages 57–66. Association for Computing Machinery.

[EBMV] Mette Agger Eriksen, Eva Brandt, Tuuli Mattelmäki, and Kirsikka Vaajakallio. Taking design games seriously: Re-connecting situated power relations of people and materials. In *Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1*, PDC '14, pages 101–110. Association for Computing Machinery.

[FGKM] Peter Fikar, Tilo Grenz, Philipp Knopp, and Tereza Maletz. Von der grounded theory zur gebauten theorie: Design games als soziologische methode.

[GCAG] Sune Gudiksen, Kirstine Christensen, Joakim Aalborg, and Klaus Greve. Explicating Stakeholder Criteria: Opening Up the Power Field through Design Games.

[HM] Zagros Hatami and Tuuli Mattelmäki. Facilitating Service Interactions with Design Games.

[KGPC] Taylor Kessner, Elisabeth Gee, and Luis Pérez-Cortés. Wicked Play: Wicked Problems, Designerly Citizens, Design Games.

[KKM] Mette Gislev Kjaersgaard, Eva Knutz, and Thomas Markussen. Design games as fieldwork: Re-visiting design games from a design anthropological perspective. 73:100994.

[NETM+] Sidsel K. Nymark Ernstsen, Christian Thuesen, Christian H. Mossing, Stig Brinck, and Anja Maier. Technology Cards: A design game for navigating in a future of digital technologies. 17(1):37–53.

[RW] Robin Roy and James P. Warren. Card-based design tools: A review and analysis of 155 card decks for designers and designing. 63:125–154.

[TPT] George Triantafyllakos, George Palaigeorgiou, and Ioannis A. Tsoukalas. Designing educational software with students through collaborative design games: The We!Design&Play framework. 56(1):227–242.

[VM] Kirsikka Vaajakallio and Tuuli Mattelmäki. Design games in codesign: As a tool, a mindset and a structure. 10(1):63–77.

[ZPZ+] Ziheng Zhang, Rui Patricio, Tengjia Zuo, Wa An, and Ruoqing Huang. Addressing cultural inertias for co-design: Exploring Chinese participants' perceptions of design games. 4:2483–2492.